

interpolated resolution, which is much higher thanks to software interpolation.

The higher the resolution, the larger is the file. In most cases, there is a trade-off between manageable file size and level of detail.

The third important parameter for a scanner is its density range. A high density range means that the scanner is able to reproduce shadow details and brightness details in one scan.

There are four common means of transferring images from scanners:

- Parallel connection through a parallel port is the slowest common transfer method. Early scanners had parallel port connections that could not transfer data faster than 70 Kbps. The primary advantage of the parallel port connection was economy—it avoided adding an interface card to the computer.
- Small Computer System Interface (SCSI), which is supported only via an additional SCSI interface card. Some SCSI scanners are supplied together with a dedicated SCSI card for a PC, although any SCSI controller can be used. During the evolution of the SCSI standard, speeds increased with backwards compatibility; a SCSI connection can transfer data at the highest speed which both the controller and the device support. SCSI has been largely replaced by USB and FireWire, either of which are directly supported by most computers, and which are easier to set up than SCSI.
- USB scanners can transfer data quickly, and are easier to use, yet cheaper than SCSI devices. The early USB 1.1 standard could transfer data at only 1.5 Mbps (slower than SCSI), but the later USB 2.0 standard can theoretically transfer up to 60 MBps (although practical rates are much lower), resulting in faster operation.
- FireWire is an interface that is much faster than USB 1.1 and comparable to USB 2.0. FireWire speeds are 25, 50 and 100 MBps